

Aditya Krishnan

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SUMMARY

Engineer and researcher (NeurIPS, STOC, ICML) with deep expertise in building scalable information retrieval systems, from search indexing to training large (multimodal) embedding and ranking models for RAG workflows.

PROFESSIONAL EXPERIENCE

Microsoft -- M365 Copilot, Semantic Search -- New York City

Mar 2025 – Present

Senior Software Engineer

- Leading evaluation of frontier technologies for the semantic index - training and evaluating state-of-the-art late-interaction multimodal embedding models (based on ColQwen) for enterprise data (achieving top performance on MTEB Vidore V3) and evaluating SLM-based agentic search models.
- Designed and implemented novel indexing algorithms that reduced DRAM and SSD footprint by over 2× across Substrate's serving fleet, delivering \$X million in annual COGS savings.
- Core contributor to [DiskANN](#) - Microsoft's open-source graph-based ANN search library built in Rust. Implementing the graph search algorithm against SSD-based persistence and optimizing algorithms using SIMD kernels.

Pinecone Systems -- New York City

Oct 2022 – Oct 2024

Senior Research Scientist

- Trained a production-grade neural reranker with a DeBERTa backbone using a novel negative sampling strategy. Managed the full ML pipeline - data curation, distributed training, hyperparameter search, and evaluation - outperforming state-of-the-art text re-rankers like Cohere-Rerank-V3 on passage retrieval benchmarks.
- Contributed to Pinecone's IVF-based serverless semantic index service built over object storage - including the serialization of indexes using Parquet, the caching service and innovations to core cluster selection algorithm that reduced data scanned during query serving by up to 50% (NeurIPS 2025).
- Built Pinecone's [1st-place solution](#) for the NeurIPS 2023 Big ANN Benchmark (OOD track) in Rust — achieving 35K+ QPS on 10M vectors with 8 vCPUs and 16 GB RAM through novel data layout and optimized SIMD kernels.

EDUCATION

Johns Hopkins University, Whiting School of Engineering -- Baltimore, MD

Sep 2018 – Sep 2022

Ph.D. in Computer Science -- Advisor: Vladimir Braverman

Thesis: Fast and Memory-Efficient Algorithms for Matrix Spectrum Approximation

Carnegie Mellon University, School of Computer Science -- Pittsburgh, PA

Aug 2013 – May 2018

B.S. and M.S. in Computer Science

TECHNICAL SKILLS

Languages and Frameworks: Rust, Python, PyTorch, Distributed Data Parallel, FSDP2

Infrastructure: Large-scale ML data pipelines and algorithms for indexing and retrieval, LLM architectures, SIMD intrinsics

SELECTED PUBLICATIONS

Authors appear in alphabetical order. Full list on Google Scholar.

NeurIPS 2025: *Optimistic Query Routing for Maximum Inner-Product Search*. With S. Bruch and F.M. Nardini

STOC 2022: *Sublinear Time Spectral Density Estimation*. With V. Braverman and C. Musco

ACML 2021: *Lifelong Learning with Sketched Structural Regularization*. With H. Li, J. Wu, S. Kolouri, P.K. Pilly, and V. Braverman

ICML 2020: *Schatten Norms in Matrix Streams: Hello Sparsity, Goodbye Dimension*. With V. Braverman, R. Krauthgamer, and R. Sinoff

HONORS & AWARDS

- NeurIPS Top Reviewer - awarded to < 10% of reviewers (2022)
- JHU MINDS TRIPODS Data Science Fellowship (2022)
- JHU CS Department Fellowship -- awarded to 2 of 50+ incoming PhD students (2018)